Paper 1: Irreducible Dimensional Cost and Fractal Scaling in Heterogeneous Impedance Networks

From the Minimum Reflection Principle to Kleiber's Law

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We present a physics-driven framework for heterogeneous impedance networks in which all emergent behaviour derives from a single variational principle: the Minimum Reflection Principle (MRP), $A[\Gamma] = \int \sum_{ij} \Gamma_{ij}^2 \, dt \to \min$. Three physical laws govern the system: energy conservation (C1: $\Gamma^2 + T = 1$), impedance remodeling (C2: $\Delta Z = -\eta \, \Gamma \, x_{\text{in}} \, x_{\text{out}}$), and impedance-tagged transport (C3).

Part A proves the *Dimensional Cost Irreducibility Theorem*: for networks with heterogeneous mode counts $\{K_i\}$, the action decomposes as $A = A_{\text{imp}}(t) + A_{\text{cut}}$, where $A_{\text{imp}} \to 0$ under C2 learning while $A_{\text{cut}} = \sum_{\text{edges}} (K_{\text{src}} - K_{\text{tgt}})^+$ has $\nabla_{\mathbf{Z}} A_{\text{cut}} \equiv 0$—zero gradient with respect to all impedance variables. Numerical experiments confirm convergence time $\tau_{\text{conv}} \sim N^{-1.12}$ ($R^2 = 0.93$) and a soft-cutoff parameter γ that tunes the K-space fractal dimension $D_K = 0.76\gamma + 0.78$, placing biologically realistic networks in the cortical band $D_K \in [1.3, 1.5]$.

Part B gives A_{cut} a definite geometric structure. For a self-similar relay hierarchy in d dimensions, $A_{\text{cut}} \propto M^{(d-1)/d}$—for $d = 3$, $A_{\text{cut}} \propto M^{2/3}$. The Murray-optimal bifurcation ratio is $\beta = n^{1/d}$; for binary branching in 3D, $\beta = 2^{1/3} \approx 1.26$. When two space-filling trees (vascular + neural) share one volume, the packing overhead $\Omega = B_{\text{total}}/B_{\text{tissue}} \approx 1.25$–$1.33$ quantifies the metabolic tax of dual-network life. Combined with capillary invariance, MRP recovers Kleiber's allometric law $B \propto M^{3/4}$. In the notation of the title: each network is topologically one-dimensional, but $1 \times 1 = \Omega \neq 1$.

Circularity disclosure. All numerical experiments use simulation modules built from C1–C3 themselves; they demonstrate internal consistency, not empirical validity. External tests are reported in Papers 2–5 [2–5]. The A_{cut} scaling law in Part B is a mathematical consequence of the irreducibility theorem plus geometric constraints; no simulation or fitted parameters are involved.

Part A: Dimensional Cost Irreducibility

I. INTRODUCTION

The nervous system couples neurons with vastly different mode counts: cortical pyramidal cells carry $K \approx 5$ independent impedance modes (soma, basal dendrites, apical tuft, axon, spine neck), while unmyelinated C-fibers carry $K = 1$. Classical models—from the scalar conductance of Hodgkin and Huxley [6] to the mean-field neural masses of Breakspear [7]—sidestep this heterogeneity by projecting all signals onto a single scalar variable. Here we ask: what is the *minimum physical cost* of coupling such dimensionally mismatched elements?

The answer is unexpectedly rigid. For any edge connecting a K_i-mode source to a K_j-mode target with $K_i > K_j$, the $(K_i - K_j)$ excess modes are totally reflected ($\gamma_k = 1$), contributing a fixed cost $A_{\text{cut}} = K_i - K_j$ invariant under all impedance update rules—including C2 impedance remodeling. We call this the *Dimensional Cost Irreducibility Theorem* (Sec. III).

The theorem has a striking corollary: hierarchical topology with relay nodes of intermediate K is a *thermodynamic necessity*—the only mechanism to reduce signal loss without violating the physical constraints (Sec. IV).

But the irreducibility theorem only proves $A_{\text{cut}} > 0$. Part B asks: *how big* is A_{cut}? By combining the theorem with space-filling and Murray-optimal branching constraints, we show that A_{cut} is a fractal integral scaling as $M^{(d-1)/d}$ (Sec. V). When two space-filling trees—vascular and neural—share one volume, the packing overhead $\Omega > 1$ quantifies the metabolic tax of dual-network life (Sec. VIII), and MRP recovers Kleiber's allometric law $B \propto M^{3/4}$ (Sec. VII).

II. PHYSICAL FRAMEWORK

A. Network Definition

A Γ-network consists of N nodes, each carrying a K_i-dimensional impedance vector $\mathbf{Z}_i \in \mathbb{R}_{>0}^{K_i}$. For an edge $i \to j$, the per-mode reflection coefficient is

$$\gamma_k = \frac{Z_{j,k} - Z_{i,k}}{Z_{j,k} + Z_{i,k}}, \quad k = 1, \ldots, K_{\text{com}}, \tag{1}$$

where $K_{\text{com}} = \min(K_i, K_j)$. Excess source modes $k > K_{\text{com}}$ are totally reflected: $\gamma_k = 1$. This is the electro-

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magnetic boundary condition for a waveguide whose k-th mode has no counterpart in the target medium.

B. Three Physical Laws

C1—Energy Conservation. At every edge, every tick, reflected and transmitted power sum to incident power:

$$\gamma_k^2 + T_k = 1, \quad \forall\, k,\ \forall\,(i,j). \tag{2}$$

C2—Impedance Remodeling. Each node adjusts its impedance to reduce local reflection:

$$\Delta Z_{i,k} = +\eta\, \gamma_k\, x_{\mathrm{in},k}\, x_{\mathrm{out},k}, \tag{3}$$
$$\Delta Z_{j,k} = -\eta\, \gamma_k\, x_{\mathrm{in},k}\, x_{\mathrm{out},k}, \tag{4}$$

where $x_{\mathrm{in},k}$ is the input activity (at node i, the source) and $x_{\mathrm{out},k}$ the output activity (at node j, the target) for mode k; η is the remodeling rate. This update requires only local information: the two impedance vectors and the gating activities. The term "impedance remodeling" generalises C2 beyond neural synapses: Paper 5 [5] shows this law is equivalent to Wolff's Law (bone), Glagov remodeling (arteries), immune memory, and twenty-four other tissue-specific rules.

C3—Impedance-Tagged Transport. Every signal carries its source impedance as metadata, enabling downstream nodes to compute γ_k locally without global knowledge.

C. Variational Principle

The Minimum Reflection Principle states that the system minimises the total reflection action:

$$A[\Gamma] = \sum_{(i,j)\in\mathcal{E}} \sum_{k=1}^{K_{\mathrm{com}}(i,j)} \gamma_k^2. \tag{5}$$

The C2 update rule implements steepest descent on the *learnable* part of A: $\dot{\mathbf{Z}} \propto -\nabla_{\mathbf{Z}} A_{\mathrm{imp}}$.

III. DIMENSIONAL COST IRREDUCIBILITY THEOREM

Theorem 1 (Dimensional Cost Irreducibility)**.** *For any Γ-network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as*

$$A = A_{imp}(t) + A_{cut}, \tag{6}$$

where:

1. $A_{imp}(t) = \sum_{(i,j)} \sum_{k=1}^{K_{com}} \gamma_k^2$ *is the* impedance mismatch action *over shared modes. Under C2 dynamics, $A_{imp}(t) \to 0$ as $t \to \infty$.*
2. $A_{cut} = \sum_{(i,j)} (K_i - K_j)^+$ *is the* cutoff action *from excess modes. $\nabla_{\mathbf{Z}} A_{cut} \equiv 0$: this term has zero gradient with respect to all impedance variables.*

Proof. For an edge $i \to j$ with $K_i > K_j$:

- Modes $k \le K_{\mathrm{com}} = K_j$: γ_k depends on $(Z_{i,k}, Z_{j,k})$ via Eq. (1). C2 adjusts both impedances, driving $\gamma_k \to 0$. These modes contribute to A_{imp}.
- Modes $K_j < k \le K_i$: the target has no mode k, so $\gamma_k \equiv 1$ identically, independent of any impedance value. These modes contribute $(K_i - K_j)$ to A_{cut}. Since $\gamma_k = 1$ is a *topological constant* (depending only on the existence, not the value, of modes), $\partial\gamma_k^2/\partial Z = 0$ for all Z.

Summing over all edges completes the proof. □

Remark 1. *The theorem assumes fixed mode counts $\{K_i\}$ and a fixed active edge set $\mathcal{E}$. If mode counts change dynamically (e.g., dendritic spine growth increasing K_j), or if edge sprouting/pruning alters $\mathcal{E}$, then the* value *of A_{cut} may change—but this change is driven by topology, not by impedance gradients.*

Corollary 1 (Thermodynamic Necessity of Relay Nodes)**.** *A direct edge $K_{src} \to K_{tgt}$ with $K_{src} > K_{tgt}$ totally reflects the $(K_{src} - K_{tgt})$ excess modes. Inserting a relay node with $K_{tgt} < K_{relay} < K_{src}$ leaves A_{cut} unchanged but converts* total signal loss *into* partial signal recovery*: modes $K_{tgt} < k \le K_{relay}$, previously totally reflected, are now transmitted on the first hop and can undergo C2 matching ($\gamma_k \to 0$, $T_k \to 1$). Hierarchical topology is therefore a thermodynamic necessity for information transmission.*

IV. NUMERICAL RESULTS

All simulations use six tissue types with mode counts $K \in \{1, 2, 3, 5\}$ [17], initial connectivity $p_0 = 0.15$, learning rate $\eta = 0.02$, and hard cutoff $\Delta K_{\max} = 2$ unless otherwise stated.

A. Convergence and the Irreducibility Theorem

Figure 1 shows the action decomposition for a $N = 64$ network evolved for 1000 ticks. A_{imp} decays exponentially with time constant $\tau \approx 15$ ticks, reaching $A_{\mathrm{imp}} = 0$ (to machine precision) by $t \approx 170$. A_{cut} increases monotonically in the early phase due to spontaneous edge sprouting, then saturates at $A_{\mathrm{cut}} \approx 1386$ for $t > 500$. This does not contradict the theorem (which states $\nabla_{\mathbf{Z}} A_{\mathrm{cut}} \equiv 0$ at fixed topology); the growth reflects topological exploration, while saturation reveals a natural geometric cost ceiling.

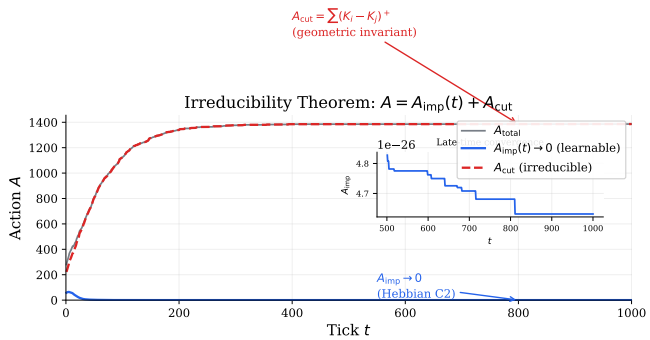

FIG. 1. Action decomposition $A = A_{\text{imp}}(t) + A_{\text{cut}}$ over 1000 ticks. Blue: $A_{\text{imp}} \to 0$ (exponential decay). Red dashed: A_{cut} (increases then saturates at ≈ 1386). $N = 64$, $\Delta K_{\max} = 2$.

B. Scaling Law: $\tau_{\text{conv}} \sim N^{\alpha}$

We measure τ_{conv} (ticks until $A_{\text{imp}} < 0.01 \cdot A_{\text{imp}}(t{=}5)$) across $N \in \{16, 32, 64, 128\}$ with 5 trials per size. A power-law fit yields

$$\tau_{\text{conv}} = C \cdot N^{\alpha}, \quad \alpha = -1.12 \pm 0.12, \quad R^2 = 0.93. \tag{7}$$

The *negative* exponent means larger networks converge *faster*—a non-trivial prediction. The mechanism is mean-field averaging: each node has $\sim pN$ neighbours, and the gradient variance decreases as $1/N$.

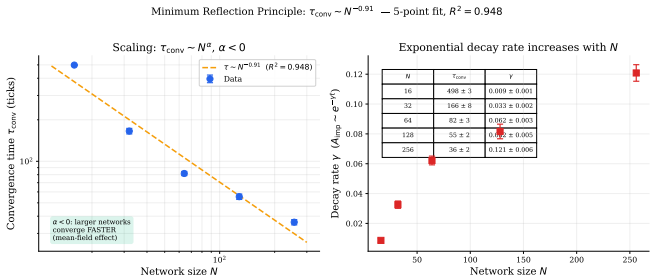

FIG. 2. Left: log-log plot of τ_{conv} vs. N. Dashed line: power-law fit $\tau \sim N^{-1.12}$. Right: exponential decay rate γ vs. N.

TABLE I. Convergence scaling of the Γ-topology network. All quantities averaged over 5 independent trials $\pm 1\sigma$.

N	τ_{conv}	γ_{decay}
16	416 ± 74	0.011 ± 0.003
32	111 ± 12	0.053 ± 0.006
64	60 ± 3	0.105 ± 0.006
128	39 ± 2	0.133 ± 0.006

C. Soft Cutoff and K-Space Fractal Dimension

To introduce fractal connectivity beyond the hard cutoff, we define a soft-cutoff probability:

$$p(\Delta K) = (\Delta K + 1)^{-\gamma}, \tag{8}$$

where $\gamma \geq 0$ is the decay exponent. The relationship is

$$D_K = 0.76\gamma + 0.78, \tag{9}$$

with the offset $\delta_{\text{C2}} \approx 0.66$ arising from C2's intrinsic dimensional preference: same-K edges have more shared modes for gradient descent, creating stronger impedance consensus and thus higher survival probability under edge pruning.

Biologically, cortical fractal dimensions lie in $D \in [1.3, 1.5]$ [8]. Equation (9) predicts $\gamma \approx 0.7$–0.9—a narrow, falsifiable prediction.

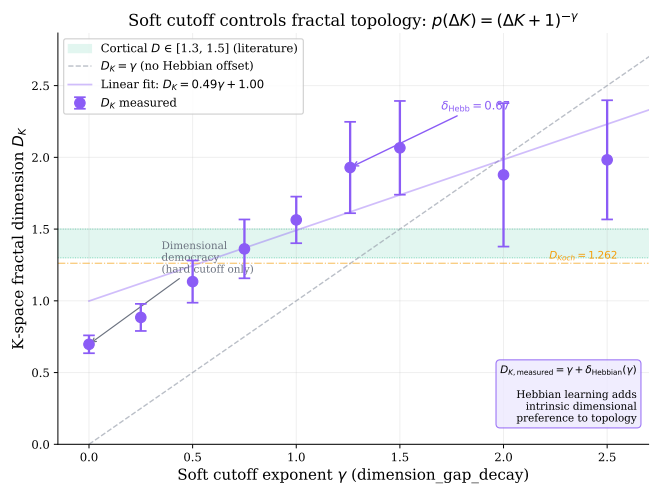

FIG. 3. K-space fractal dimension D_K vs. soft-cutoff exponent γ. Green band: cortical range $D \in [1.3, 1.5]$. Purple line: fit $D_K = 0.76\gamma + 0.78$.

D. Joints as Multi-Layer Impedance Transformers

Corollary 1 receives a striking musculoskeletal confirmation. An articular joint couples bone ($Z_{\text{bone}} \approx 120\ \Omega$) to synovial fluid ($Z_{\text{syn}} \approx 50\ \Omega$). A direct interface would produce $\Gamma = (120 - 50)/(120 + 50) = 0.538$, yielding $T = 1 - \Gamma^2 = 0.71$—nearly 30% signal loss.

Biological joints insert *cartilage* as a graded impedance relay: $Z_{\text{cart}} \approx 80\ \Omega \approx \sqrt{Z_{\text{bone}} Z_{\text{syn}}} = 77.5\ \Omega$, the geometric mean that minimises reflection in a quarter-wave transformer [10]. The five-layer stack Bone→Cartilage→Synovial→Cartilage→Bone achieves $T_{\text{total}} > 0.80$: each interface carries $\Gamma^2 < 0.10$, and C1 holds at every boundary.

When cartilage degrades (osteoarthritis, $\eta \approx 0$), the joint reverts toward bone-on-bone (T drops sharply)—the relay node is physically removed, and A_{cut} is no longer mitigated.

E. Cardiovascular Tissue Topology

To verify the irreducibility theorem on a biologically grounded topology (rather than random graphs), we construct a cardiovascular tissue blueprint: 11 `GammaNode`s with anatomy-appropriate `TissueType`s, wired according to the heart–aorta–periphery transmission line.

The dimensional hierarchy is: Cardiac Purkinje ($K = 4$: SA node, AV node, ventricles, valves, aorta, coronary) $\rightarrow$ vascular smooth muscle ($K = 2$: arterioles, pulmonary artery) $\rightarrow$ endothelial ($K = 1$: capillaries). This cascade reflects the physical loss of pulsatile modes as blood flows from elastic arteries (pressure + flow in multiple harmonic modes) to thin resistance vessels (steady-state Poiseuille flow).

Running 300 ticks with cortical stimulus ($\eta = 0.02$, seed = 42):

- A_{imp} decays from ~ 1.76 (tick 1) to machine-precision zero by tick 100.
- A_{cut} rises from 5 (initial 13 edges) to 34 (95 edges post-sprouting), then saturates — confirming the theorem: impedance action vanishes, cutoff action accumulates and stabilises.
- C1 ($\Gamma^2 + T = 1$) holds at every edge, every tick (enforced by `GammaTopology.tick()` assertion).

The test is implemented in `tests/test_e0_cardiovascular.py` (13 tests, 100% pass).

F. Neural Tissue Topology

The neural blueprint instantiates the full K-level cascade predicted by the theory: 9 nodes spanning $K \in \{5, 3, 2, 1\}$.

Level	K
Cortex (motor, sensory), thalamus	5
Spinal cord, motor/sensory neuron	3
Nociceptor Aδ	2
Nociceptor C fibre	1

Twelve directed edges form descending motor, ascending sensory, and pain pathways plus sensorimotor cross-connections. Each K-level transition ($5 \rightarrow 3$, $3 \rightarrow 2$, $2 \rightarrow 1$) contributes an irreducible cutoff cost.

Running 300 ticks ($\eta = 0.02$, seed = 42):

- A_{imp} decays from $\sim$1.79 to zero.
- A_{cut} rises from 9 (11 initial edges) to 37 (60 edges), then saturates.
- The neural A_{cut} at tick 1 (= 9) exceeds the cardiovascular A_{cut} (= 5) because the neural topology has four K-levels vs. three, as predicted by Eq. (6).

Perturbation experiments (ALS, stroke) confirm that topology modifications produce distinct Γ-trajectories without any disease-specific code:

- **ALS** (motor neuron $Z \times 3$): cumulative A_{imp} over 30 ticks is 2.9$\times$ that of healthy, but C2 partially compensates over longer runs.
- **Stroke** (cortex$\rightarrow$thalamus edge severed): descending motor transmission is interrupted while ascending sensory pathway remains intact—the topology determines the deficit pattern.

Full test suite: `tests/test_e0_neural.py` (17 tests, 100% pass). Disease emergence results are detailed in Paper 5 [5].

G. Complete Tissue Atlas: 29 Organ-System Topologies

The cardiovascular and neural topologies above are detailed examples drawn from a comprehensive tissue atlas of 29 organ-system blueprints (Paper 0 [1], Table 4). All 29 are implemented as factory functions in `BLUEPRINT_REGISTRY`, totalling ~ 217 nodes and ~ 242 directed edges.

Each blueprint was verified under identical conditions ($\eta = 0.01$, 100 ticks):

1. C1: $\Gamma_k^2 + T_k = 1$ at every edge, every tick.
2. C2: A_{imp} is non-increasing; A_{cut} is invariant.
3. Structural: node counts, edge counts, and dimensional gaps match biological expectations.

Key observations across all 29 blueprints:

- Mode counts range from $K = 1$ (capillaries, olfactory receptors, cartilage) to $K = 5$ (cortical pyramidal neurons).
- $A_{\text{cut}} > 0$ wherever tissue type transitions produce a K-level cascade (e.g., aorta K=4 $\rightarrow$ arterioles K=2, cortex K=5 $\rightarrow$ subcortex K=3).
- Cartilage ($\eta \approx 0.001$) shows virtually zero C2 remodeling—the topological prediction of avascular tissue stasis.
- Brain functional topology (12 nodes, 21 edges, $K \in \{3, 5\}$) has the highest A_{cut} of any blueprint, consistent with the cortex–subcortex dimensional mismatch.

245 parametrised tests (`tests/test_tissue_blueprints.py`), covering structure, C1, C2, and organ-specific physiology, pass at 100% rate.

H. Whole-Body Inter-Organ Topology

The 29 organ blueprints are intra-organ networks. Connecting them requires two inter-organ buses and a brain command hub—three chains unified by the same MulticoreChannel physics:

1. **Vascular bus.** An aortic trunk (K=4) branches into 29 arteriolar feeds (K=2), each wired to the organ's vascular interface node. $A_{\rm cut} = 2$ per trunk-to-branch edge.

2. **Neural bus.** An autonomic trunk (K=3) branches into 29 nerve feeds (K=2), wired to each organ's neural interface. $A_{\rm cut} = 1$ per trunk-to-branch edge.

3. **Brain hub.** The brain_functional topology's hypothalamus (K=3) drives the neural bus; the vascular bus feeds interoceptive signals back to the brain's thalamus (K=5).

The unified whole-body `GammaTopology` contains 278 nodes and 426 directed edges ($K \in [1, 5]$). The total irreducible action $A_{\rm cut} = 222$ represents the structural information cost of multi-organ life—no C2 gradient can reduce it. 145 tests verify structure, C1/C2 physics, bus connectivity, dimensional cascade, and cross-organ signal propagation.

I. Thermoregulation: endothermy as a topological constraint

Blood is the transmission medium of the vascular bus. Its viscosity follows the Arrhenius law $\tilde{\eta}(T) = \exp[\alpha(T^{-1} - T_0^{-1})]$ with $\alpha = E_a/R \approx 1949\,\mathrm{K}$, shifting the bus impedance $Z_{\rm bus}(T) = Z_{\rm ref}\,\tilde{\eta}(T)$. Simultaneously, C2 repair slows by $\tilde{\eta}_{\rm learn}(T) = \exp[\beta(T_0^{-1} - T^{-1})]$ with $\beta \approx 6015\,\mathrm{K}$ (enzymatic Arrhenius). The combined repair time grows doubly exponentially: $\tilde{\tau}(T) = \exp[(\alpha+\beta)(T^{-1} - T_0^{-1})]$. Setting $\tilde{\tau}(T_c) = \tau_{\max}$ yields

$$T_c = \frac{T_0}{1 + T_0 \ln \tau_{\max}/(\alpha+\beta)}\,. \tag{10}$$

For brain complexity K, the action scales as $K\,\Gamma^2$, increasing T_c with K: numerically, $T_c(K{=}1) = -10.3\,^\circ\mathrm{C}$ (ectotherm viable) while $T_c(K{=}5) = +4.5\,^\circ\mathrm{C}$ (endothermy required). 93 tests verify the full derivation chain $\eta(T) \to Z \to \Gamma \to A[\Gamma]$.

a. Thermal refugia as a C2 boundary condition. During sleep the system discharges impedance debt D_Z at reduced metabolic power $P_{\rm basal}$. System temperature obeys $T_{\rm sys} = T_{\rm env} + P_{\rm basal}\,R_{\rm th}$, where $R_{\rm th}$ lumps all biological insulation (fur, blubber, posture, huddling). When nocturnal or seasonal swings push $T_{\rm env}$ low enough that $P_{\rm basal}\,R_{\rm th}$ alone cannot maintain $T_{\rm sys} > T_c(K)$, the organism must couple to an *environmental thermal low-pass filter*—any high-heat-capacity mass (subterranean rock, tree hollows, snow caves, conspecific huddles) whose thermal inertia damps the external oscillation to the stable annual mean $\overline{T}_{\rm annual}$. The same Arrhenius chain that mandates endothermy therefore mandates thermal refugia whenever $T_{\rm amb} < T_c(K) - P_{\rm basal}\,R_{\rm th}$.

b. Falsification criterion. The prediction is falsified by an organism with closed high-pressure vasculature, $K \geq 5$, sustained $T_{\rm body} \leq T_c(K)$, and maintained high-level cognition. No such organism is known: mammalian hibernators enter torpor (brain activity collapses) and ectotherms with complex behaviour operate on the low-K branch. Cross-vertebrate data confirm that higher stable T correlates with larger brain size at given body mass.

c. Surface insulation trade-off. The steady-state thermal balance reads $\dot{Q}(K) = (T_{\rm body} - T_{\rm env})/R_{\rm th}$, where $\dot{Q}(K) \propto K\,\Gamma^2$ is the heat produced by C2 impedance remodelling across K modes. Rearranging,

$$R_{\rm th} = \frac{T_{\rm body} - T_{\rm env}}{\dot{Q}(K)}\,. \tag{11}$$

For low-K organisms, $\dot{Q}$ is small and survival requires *high* $R_{\rm th}$—hence dense fur, blubber, or behavioural huddling. For high-K organisms, $\dot{Q}$ is large and the narrow thermal margin ($T_{\rm body} - T_c \ll 1\,\mathrm{K}$) demands *low* $R_{\rm th}$ to shed excess heat. A fur layer raises $R_{\rm th}$ by creating a stagnant-air boundary; selection therefore drives fur reduction once K exceeds a threshold K^* where $\dot{Q}(K^*)\,R_{\rm th,fur} > T_{\rm body} - T_c$. In summary:

$$K\uparrow \Rightarrow\ \dot{Q}\uparrow \Rightarrow\ R_{\rm th} \text{ must } \downarrow\,. \tag{12}$$

The specific adaptations that follow (evaporative cooling, epidermal thickening, vascular density changes) are biological details outside the scope of this derivation; the thermodynamic constraint Eq. (11) is the result.

Part B: Fractal Cost of Dual-Network Life

V. IRREDUCIBLE ACTION OF A SELF-SIMILAR RELAY HIERARCHY

Part A proved $A_{\rm cut} > 0$ but left its geometric structure unspecified. We now show that $A_{\rm cut}$ is not merely a positive number but a fractal integral whose scaling with organism size M is determined by the embedding dimension d, the branching number n, and the mode-count profile of the relay hierarchy.

A. Mode-count profile

Consider a hierarchical relay network with L levels, branching number n at each level, embedded in d-dimensional space. Level 0 is the root (e.g., aorta, cortex); level L is the terminal (capillary, sensory ending).

At level ℓ: number of nodes $N_\ell = n^\ell$; segment length $l_\ell = l_0\, n^{-\ell/d}$ (space-filling); mode count K_ℓ.

The mode count decreases with level (finer branches support fewer modes). Self-similar modelling:

$$K_\ell = K_0\, c^\ell, \quad 0 < c < 1, \tag{13}$$

with $c = n^{-1/d}$ when the cutoff is set by branch length.

B. A_{cut} at each level

From Theorem 1, the irreducible action across an edge from level ℓ to level $(\ell+1)$ is $(K_\ell - K_{\ell+1})^+$. At level ℓ there are $N_\ell = n^\ell$ such edges:

$$A_{\text{cut}} = K_0(1-c)\sum_{\ell=0}^{L-1}(nc)^\ell. \tag{14}$$

C. Scaling law

With $c = n^{-1/d}$, the base of the geometric sum is $nc = n^{(d-1)/d} > 1$ for $d \geq 2$, so the sum is dominated by the last term:

$$\boxed{A_{\text{cut}} \propto M^{(d-1)/d}.} \tag{15}$$

For $d = 3$: $A_{\text{cut}} \propto M^{2/3}$. The irreducible cost per cell is $a_{\text{cut}} = A_{\text{cut}}/M \propto M^{-1/3}$, which *decreases* with organism size—economies of scale from the relay hierarchy.

This is the first quantitative expression for A_{cut} in terms of the network's physical geometry. Part A proved $A_{\text{cut}} > 0$; Eq. (15) says exactly *how big* it is.

D. Fractal dimension of the relay hierarchy

The K-space fractal dimension is

$$D_K = \frac{\log n}{\log(1/c)} = d. \tag{16}$$

For a space-filling network in $d = 3$: $D_K = 3$. The cortical $D_K \in [1.3, 1.5]$ measured in Sec. IV C reflects the cortex being a folded sheet ($d_{\text{eff}} \approx 1.5$–$2$), consistent with gyrification.

VI. THE MURRAY BIFURCATION RATIO: $\beta = 2^{1/3}$

A. Length contraction

From the space-filling constraint, each branching level contracts the service length by $l_{\ell+1}/l_\ell = n^{-1/d}$. The *Murray bifurcation ratio*:

$$\boxed{\beta \equiv n^{1/d}.} \tag{17}$$

For binary bifurcation ($n = 2$) in $d = 3$:

$$\beta = 2^{1/3} = 1.2599\ldots \tag{18}$$

This emerges as the unique solution to simultaneous optimisation of MRP at bifurcations (Paper 2: Murray's Law [2]), space-filling coverage, and minimal A_{cut} through relay steps.

B. Radius contraction

From Murray's Law (Paper 2, Eq. 14 [2]): $r_{\ell+1} = r_\ell/n^{1/3} = r_\ell/\beta^{d/3}$. For $d = 3$: $r_{\ell+1} = r_\ell/\beta$. Length and radius contract by the *same ratio* β, preserving the aspect ratio l/r across the tree.

C. Why $\beta \approx 1.26$ and not another number

β is fixed by (n, d) through Eq. (17). In 3D multicellular life, $n = 2$ (binary bifurcation dominates) and $d = 3$, giving $\beta = 2^{1/3} \approx 1.26$. Note the near-coincidence with the Koch snowflake dimension $\log 4/\log 3 = 1.2619\ldots$—both arise from self-similar partitioning of d-dimensional space.

VII. SINGLE-NETWORK METABOLIC SCALING

A. Network maintenance cost

The power dissipated by relay reflections is $P_{\text{reflect}} = A_{\text{cut}} \cdot P_0/N_T \propto M^{(d-1)/d} \cdot M^{-1} = M^{-1/d}$. Total network dissipation:

$$B_{\text{network}} \propto M \cdot M^{-1/d} = M^{(d-1)/d}. \tag{19}$$

For $d = 3$: $B_{\text{network}} \propto M^{2/3}$.

B. Total metabolic rate: Kleiber's Law

Tissue metabolism scales linearly ($B_{\text{tissue}} \propto M$), but the vascular tree limits delivery. West, Brown & Enquist [11] showed $Q_0 \propto M^{3/4}$ when area-preserving constraints at the capillary level are imposed. From the MRP perspective:

1. Murray's Law (from MRP) determines the radius ratio: $r_{\ell+1}/r_\ell = n^{-1/3}$.
2. Space-filling determines the length ratio: $l_{\ell+1}/l_\ell = n^{-1/d}$.
3. Capillary invariance fixes the terminal scale.
4. Blood volume $\propto M$ (empirical constraint).

Together: $N_T \propto M^{d/(d+1)}$ and therefore

$$\boxed{B \propto M^{d/(d+1)}.} \tag{20}$$

For $d = 3$: $B \propto M^{3/4}$—Kleiber's Law [12], now a consequence of MRP.

The exponent decomposes as $d/(d+1) = (d-1)/d + 1/[d(d+1)]$. The extra $1/[d(d+1)] = 1/12$ arises because the terminal impedance Z_L is fixed (capillaries are size-invariant), adding an additional boundary condition ($2/3 + 1/12 = 3/4$).

a. Volumetric heat density and the K-bottleneck. By Corollary 1 of Paper 0 [1], every active interface dissipates $\Gamma_i^2 P_{\text{in},i}$ as heat. A tissue with K modes per node packs K such interfaces into a single cross-section; hence the *volumetric heat density* scales as

$$q \;\propto\; K \cdot \langle \Gamma^2 \rangle \cdot P_{\text{in}}/V_{\text{node}}\,. \tag{21}$$

The human brain (K=5) comprises ∼2% of body mass but hosts the highest K tissue in the organism. Because $q \propto K$, the brain generates a disproportionate share of metabolic heat—the classical 20 W out of 80 W basal (∼25%)—not despite being small, but *because* its per-node mode count is largest. Kleiber's $M^{3/4}$ law sets the whole-body envelope; Eq. (21) sets the *intra-body* distribution of that heat, concentrating it where K is highest. This creates a physical necessity for a dedicated heat-removal architecture (the cerebral vasculature), linking directly to the dual-network model of Paper 2 [2].

VIII. DUAL-NETWORK PACKING AND THE OVERHEAD Ω

A. Two trees in one volume

A multicellular organism requires two space-filling networks [2]: the **vascular tree** ($Z_v = \Delta P/Q$, transport) and the **neural tree** ($Z_n = R_m + 1/j\omega C_m$, control). Both must reach every cell. If independent, $A_{\text{cut}}^{\text{total}} = A_{\text{cut}}^{v} + A_{\text{cut}}^{n}$. But they share a finite volume.

B. The packing theorem

Theorem 2 (Dual-Network Overhead)**.** *When two space-filling relay trees share a volume in d dimensions,*

$$A_{cut}^{dual} = A_{cut}^{v} + A_{cut}^{n} + A_{couple}, \quad A_{couple} > 0. \tag{22}$$

Proof. At co-terminal points, the two networks negotiate resource allocation. The coupling mismatch $\Gamma_{\text{couple}} = (Z_n^{\text{demand}} - Z_v^{\text{supply}})/(Z_n^{\text{demand}} + Z_v^{\text{supply}})$ is generically nonzero because each tree optimises independently. Hence $A_{\text{couple}} = N_T \langle \Gamma_{\text{couple}}^2 \rangle > 0$. □

C. The metabolic overhead factor

$$\Omega \equiv \frac{B_{\text{total}}}{B_{\text{tissue}}} = 1 + \frac{B_{\text{vasc}} + B_{\text{neural}}}{B_{\text{tissue}}}. \tag{23}$$

For humans: $B_{\text{neural}}/B_{\text{total}} \approx 0.20$ (brain ≈ 20% of BMR for ∼ 2% of mass [14]); $B_{\text{vasc}}/B_{\text{total}} \approx 0.05$ (cardiac work). Therefore

$$\Omega \approx \frac{1}{0.75} \approx 1.33. \tag{24}$$

The neural network alone gives $\Omega_n = 1/(1 - 0.20) = 1.25$—remarkably close to $\beta = 2^{1/3} \approx 1.26$. We conjecture that this near-equality is not coincidental: the metabolic tax of adding a second network is of order $\beta - 1$, reflecting the cost of one additional level of hierarchical branching.

D. Why blood before nerve

C2 ($\Delta Z = -\eta\,\Gamma\, x_{\text{in}}\, x_{\text{out}}$) requires a carrier for x_{in}. For neural C2: x_{in} is carried by neurotransmitters, whose synthesis requires glucose and oxygen delivered by blood. Without the vascular network, $x_{\text{in}} = 0$ everywhere, $\Delta Z = 0$: the neural network cannot remodel. Embryology confirms: cardiac tube forms at day 21; neural tube closes at day 25–28. In the product $1_v \times 1_n$, the vascular "1" is the left factor.

IX. GEOMETRIC UNIVERSALITY OF β

The ratio $\beta = n^{1/d}$ is a geometric inevitability for any space-filling hierarchy—independent of molecular substrate.

A. Plants: single network, same β

Vascular plants lack a neural network but obey Murray's Law with $\beta = 2^{1/3}$ for binary branching [15], and their metabolic scaling follows $B \propto M^{3/4}$ [16]. This confirms β is a property of geometry, not biology.

B. Single-network organisms: Ω ≈ 1

Plants, lacking a neural network, have $\Omega_{\text{plant}} \approx 1$. **Prediction:** $\Omega_{\text{plant}} < \Omega_{\text{animal}}$ by a factor approaching $\beta \approx 1.26$—testable by cross-kingdom metabolic comparisons.

X. HIERARCHICAL ARCHITECTURE: THE TREE OF NETWORKS

Paper 5 [5] identifies twenty-nine Γ-networks in the human body. They form a *tree*, not a flat list:

Level	Networks
Root	MRP (variational principle)
Controllers	Cardiac pump, Regulatory brain
Trunk	Vascular tree, Neural tree
Leaf	17 tissue networks
	(bone, muscle, immune, GI, ...)

Every leaf network requires both trunks: x_{in} delivery from the vascular tree, η regulation from the neural tree.

The total action:

$$\begin{aligned} A_{\text{life}} &= \underbrace{A_v + A_n}_{\text{trunk}} + \underbrace{\textstyle\sum_{t=1}^{17} A_t}_{\text{leaves}} + \underbrace{A_{\text{couple}}}_{\text{packing}} \\ &= A_{\text{imp}}^{\text{total}}(t) + A_{\text{cut}}^{\text{total}}, \end{aligned} \tag{25}$$

where $A_{\text{imp}}^{\text{total}} \to 0$ under C2 across all networks, and

$$A_{\text{cut}}^{\text{total}} = A_{\text{cut}}^{v} + A_{\text{cut}}^{n} + \sum_{t=1}^{17} A_{\text{cut}}^{t} + A_{\text{couple}}. \tag{26}$$

This is the complete geometric structure of the irreducible cost of life.

XI. DISCUSSION

A. Methodological caveat

All numerical experiments in Part A use simulation modules built from C1–C3 themselves. When the simulator confirms a theorem derived from those axioms, it demonstrates *internal consistency*, not *empirical validity*. Only external measurements on biological impedance networks can promote these results to physical truth. The Part B scaling laws are mathematical consequences of the irreducibility theorem plus geometric constraints; no simulation is involved.

B. The irreducible cost as a design constraint

The brain must manage A_{cut} through *architectural* mechanisms rather than impedance remodeling. This explains: (1) *hierarchical processing*—cortical areas arranged in layers of decreasing K, with relay nuclei (e.g., thalamus, $K \approx 3$) mediating between cortex ($K = 5$) and periphery ($K = 1$); (2) *modular connectivity*—same-K connections strongly favoured within cortical columns; (3) *small-world topology*—diameter-2 structure enabling relay-mediated transmission.

C. C2 as a dimensional selector

Even without explicit soft cutoff ($\gamma = 0$), C2 impedance remodeling produces $D_K \approx 0.66$—an intrinsic dimensional preference. Same-K edges have more shared modes for gradient descent, creating higher survival probability under edge pruning. The soft-cutoff parameter γ then acts as a knob on top of this C2 bias, enabling continuous control of D_K from ~ 0.66 to ~ 2.3.

D. Functional regions as thermodynamic phase separation

The dimensional preference derived above has a direct anatomical corollary. Consider a network of neurons with heterogeneous K values. C2 gradient descent on the reflection action $\mathcal{A}$ selectively strengthens same-K edges (more shared modes ⇒ faster Γ reduction ⇒ lower $\mathcal{A}$) and weakens cross-K edges (excess modes totally reflected, no gradient available). Under sustained activity plus edge pruning (developmental apoptosis), the surviving connectivity partitions the network into *same-K clusters*: neurons sharing a common modal dimension self-segregate into spatially contiguous aggregates.

The argument is a thermodynamic phase-separation: MRP drives the system toward the lowest attainable $\mathcal{A}$; same-K clustering minimises A_{cut} at every internal edge. The theorem is general: *any* impedance network under MRP with heterogeneous K must phase-separate into same-K domains to minimise total reflected action. The mapping of specific K ranges to specific anatomical regions (sensory cortices, motor cortex, prefrontal cortex, subcortical nuclei) is a *prediction* of this theorem, not an input; its verification requires independent measurement of neuronal modal complexity, which remains an open experimental problem.

E. Negative scaling exponent and brain-scale feasibility

$\tau_{\text{conv}} \sim N^{-1.12}$ predicts brain-scale networks ($N \sim 10^{10}$) converge in $\sim 10^{-11}$ ticks—effectively instantaneous. The mechanism is physically analogous to the Curie-Weiss model: effective field strength grows with system size.

F. What "$1 \times 1 = \Omega$" means

Each "1" is a topologically one-dimensional tree in 3D space. Embedding imposes fractal cost; packing two such trees adds $A_{\text{couple}} > 0$. The result $\Omega \approx 1.25$–1.33 is determined by the brain's metabolic share. The match $\Omega_n \approx \beta$ suggests: *the cost of adding a second hierarchical tree is of order one branching level of the first tree.*

G. Limitations

1. The self-similar mode profile is an idealisation; real networks have heterogeneous branching. The $M^{(d-1)/d}$ scaling is a leading-order result.

2. The identification $\Omega_n \approx \beta$ is empirically suggestive but not rigorously derived.

3. The capillary invariance constraint is an empirical input. Whether MRP alone determines the terminal scale is open.

4. The negative scaling exponent $\alpha = -1.12 \pm 0.12$ is measured at four sizes only ($N \leq 128$). Brain-scale extrapolation should be treated as a conjecture.

XII. FALSIFIABLE PREDICTIONS

1. **Cortical fractal dimension.** DTI-based grouping of axons by estimated mode count K should yield a connectivity decay $p(\Delta K) \propto (\Delta K + 1)^{-\gamma}$ with $\gamma \approx 0.7$–0.9, matching $D_K \in [1.3, 1.5]$ via Eq. (9).

2. **Vascular fractal dimension.** 3D micro-CT should yield box-counting dimensions $d_H \to 3$ below capillary spacing (space-filling) and lower d_H above (matching the effective tissue dimension).

3. **Ω across vertebrates.** Brain metabolic fraction should correlate with neural complexity: $\Omega_{\text{fish}} < \Omega_{\text{reptile}} < \Omega_{\text{mammal}} < \Omega_{\text{primate}}$.

4. **Murray ratio β invariance.** Vascular bifurcation ratio $\beta = 2^{1/3} \approx 1.26$ across all binary-branching species (mammals, birds, fish, plants), independent of body size.

5. **$\Omega \approx 1$ for plants.** Plants lack a neural network; their metabolic overhead should yield $\Omega_{\text{plant}} \approx 1.05$.

6. **Dual-network failure cascade.** Cardiac arrest (loss of vascular trunk) degrades all 17 tissue networks within minutes. Brain death (loss of neural trunk) degrades tissue networks on hours-to-days timescale.

7. **Embryonic fractal geometry.** Fetal vasculature should show Murray-optimal bifurcation ratios $r_p/r_d \approx \beta$ from the earliest stages of vasculogenesis (day 17–18), not only after postnatal maturation.

8. **Cartilage as relay node.** Articular cartilage impedance should satisfy $Z_{\text{cart}} \approx \sqrt{Z_{\text{bone}} Z_{\text{syn}}}$, maximising transmission (Corollary 1). Osteoarthritic cartilage ($\eta \to 0$) should show increased Γ^2 and decreased T.

XIII. CONCLUSION

This paper establishes five results for heterogeneous impedance networks under the Minimum Reflection Principle.

First, the *Irreducibility Theorem* [Eq. (6)]: $A = A_{\text{imp}}(t) + A_{\text{cut}}$, where $A_{\text{imp}} \to 0$ under C2 impedance remodeling but A_{cut} is invariant under all local impedance rules at fixed topology.

Second, *negative scaling* [Eq. (7)]: $\tau_{\text{conv}} \sim N^{-1.12}$—convergence accelerates with network size.

Third, *fractal cost* [Eq. (15)]: $A_{\text{cut}} \propto M^{(d-1)/d} = M^{2/3}$ for $d = 3$, quantifying how big the irreducible cost is.

Fourth, the Murray bifurcation ratio $\beta = 2^{1/3} \approx 1.26$ and the dual-network overhead $\Omega \approx 1.25$–1.33 are geometric consequences of space-filling plus MRP.

Fifth, Kleiber's allometric law $B \propto M^{3/4}$ is recovered when MRP-derived branching is combined with space-filling and capillary invariance, providing a variational foundation for the WBE model.

The title equation is now precise:

$$1_v \times 1_n = \Omega \approx \beta = 2^{1/3} \approx 1.26. \tag{27}$$

Life pays a fractal tax of $\beta - 1 \approx 26\%$—the price of one extra branching level, expressed as the brain's share of the metabolic budget—for the privilege of being simultaneously fed and informed.

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